

BPU Examples

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BHT Example

Considering a 2-bit saturating counter BHT of 1K entries; and assuming that the processor executes the following code fragment, determine the BHT final state and calculate the misprediction ratio for the presented case.

```
for (i=1; i<=100; i++) {  
    read_data();  
    if (aa==2)  
        aa = 0;  
    if (bb==2)  
        bb = 0;  
    if (aa == bb)  
    {...}  
}
```

BHT Example

Considering a 2-bit saturating counter BHT of 1K entries; and assuming that the processor executes the following code fragment, determine the BHT final state and calculate the misprediction ratio for the presented case.

```
for(i=1;i<=100;i++){  
    read_data();  
    if (aa==2)  
        aa = 0;  
    if (bb==2)  
        bb = 0;  
    if (aa == bb)  
    {...}  
}  
  
L0:    ...  
        DADDUI R3, R1, #-2  
        BNEZ R3, L1  
        DADD R1, R0, R0  
L1:    DADDUI R3, R2, #-2  
        BNEZ R3, L2  
        DADD R2, R0, R0  
L2:    DSUB R3, R1, R2  
        BNEZ R3, L3  
        ...  
L3:    ...  
        DADDI R4, R4, #-1  
        BNEZ R4, L0  
        ...
```

BHT Example

Considering a 2-bit saturating counter BHT of 1K entries; and assuming that the processor executes the following code fragment, determine the BHT final state and calculate the misprediction ratio for the presented case.

Case 1:

-Program inputs for aa and bb are always different than 2.

Iteration 1

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	0	NT	
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	0	NT	
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	0	NT	
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	0	NT	
0x0040		...	0	NT	

Iteration 1

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	1	NT	1
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	0	NT	
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	0	NT	
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	0	NT	
0x0040		...	0	NT	

Iteration 1

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	1	NT	1
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	1	NT	1
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	0	NT	
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	0	NT	
0x0040		...	0	NT	

Iteration 1

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	1	NT	1
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	1	NT	1
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	1	NT	1
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	1	NT	1
0x0040		...	0	NT	

Iteration 2

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	2	NT	2
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	2	NT	2
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	2	NT	2
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	2	NT	2
0x0040		...	0	NT	

Iteration 3

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	3	T	2
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	3	T	2
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	3	T	2
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	3	T	2
0x0040		...	0	NT	

Iteration 4

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	3	T	2
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	3	T	2
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	3	T	2
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	3	T	2
0x0040		...	0	NT	

Iteration 100

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	3	T	2
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	3	T	2
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	3	T	2
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	2	T	3
0x0040		...	0	NT	

BHT Example

Considering a 2-bit saturating counter BHT of 1K entries; and assuming that the processor executes the following code fragment, determine the BHT final state and calculate the misprediction ratio for the presented case.

Case 1:

-Program inputs for aa and bb are always different than 2.

Mispredition ratio =

Number of mispredicted branches / Total branches

Mispredition ratio = 9 / 400

Mispredition ratio = 2.25%

BHT Example - Case 2

Considering a 2-bit saturating counter BHT of 1K entries; and assuming that the processor executes the following code fragment, determine the BHT final state and calculate the misprediction ratio for the presented case.

Case 2:

-Program inputs for aa and bb alternate values different than 2 and equal to 2.

Case 2 - Iteration 1

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	0	NT	
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	0	NT	
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	0	NT	
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	0	NT	
0x0040		...	0	NT	

Case 2 - Iteration 1

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	1	NT	1
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	1	NT	1
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	1	NT	1
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	1	NT	1
0x0040		...	0	NT	

Case 2 - Iteration 2

aa = 2
bb = 2

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	0	NT	1
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	0	NT	1
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	0	NT	1
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	2	NT	2
0x0040		...	0	NT	

Case 2 - Iteration 3

aa = 4
bb = 3

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	1	NT	2
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	1	NT	2
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	1	NT	2
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	3	T	2
0x0040		...	0	NT	

Case 2 - Iteration 4

aa = 2
bb = 2

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	0	NT	2
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	0	NT	2
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	0	NT	2
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	3	T	2
0x0040		...	0	NT	

Case 2 - Iteration 100

aa = 2
bb = 2

address	Instruction		BHT	Prediction	misP. counter
0x0000	L0:	...	0	NT	
...		...	0	NT	
0x0010		DADDUI R3, R1, #-2	0	NT	
0x0014		BNEZ R3, L1	0	NT	50
0x0018		DADD R1, R0, R0	0	NT	
0x001C	L1:	DADDUI R3, R2, #-2	0	NT	
0x0020		BNEZ R3, L2	0	NT	50
0x0024		DADD R2, R0, R0	0	NT	
0x0028	L2:	DSUB R3, R1, R2	0	NT	
0x002C		BNEZ R3, L3	0	NT	50
0x0030		...	0	NT	
0x0034	L3:	...	0	NT	
0x0038		DADDI R4, R4, #-1	0	NT	
0x003C		BNEZ R4, L0	2	T	3
0x0040		...	0	NT	

BHT Example - Case 2

Considering a 2-bit saturating counter BHT of 1K entries; and assuming that the processor executes the following code fragment, determine the BHT final state and calculate the misprediction ratio for the presented case.

Case 2:

-Program inputs for aa and bb alternate values different than 2 and equal to 2.

Mispredition ratio =

Number of mispredicted branches / Total branches

Mispredition ratio = 153 / 400

Mispredition ratio = 38.25%

(2,2) Example

Considering a (2,2) correlating predictor of 1K entries; and assuming that the processor executes the following code fragment, determine the (2,2) final state and calculate the misprediction ratio for the presented case.

	L0:	...
for(i=1;i<=100;i++){		DADDUI R3, R1, #-2
read_data();		BNEZ R3, L1
if (aa==2)		DADD R1, R0, R0
aa = 0;	L1:	DADDUI R3, R2, #-2
if (bb==2)		BNEZ R3, L2
bb = 0;	L2:	DADD R2, R0, R0
if (aa == bb)		DSUB R3, R1, R2
{...}		BNEZ R3, L3
}	L3:	...
		...
		DADDI R4, R4, #-1
		BNEZ R4, L0
		...

(2,2) Example

Considering a (2,2) correlating predictor of 1K entries; and assuming that the processor executes the following code fragment, determine the (2,2) final state and calculate the misprediction ratio for the presented case.

Case 1:

-Program inputs for aa and bb alternate values different than 2 and equal to 2.

(2,2) - Iteration 1

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	00	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	0	0	0	0		
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	0	0	0		
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	0		
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	0	0	0	0		
0x0040	...	0	0	0	0		

(2,2) - Iteration 1

aa = 3
bb = 4

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	00	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	0	0	0	01	1
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	0	0	0		
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	0		
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	0	0	0	0		
0x0040	...	0	0	0	0		

(2,2) - Iteration 1

aa = 3
bb = 4

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	00	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	0	0	0	01	1
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	1	0	0	11	1
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	0		
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	0	0	0	0		
0x0040	...	0	0	0	0		

(2,2) - Iteration 1

aa = 3
bb = 4

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	00	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	0	0	0	01	1
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	1	0	0	11	1
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	1	11	1
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	0	0	0	1	11	1
0x0040	...	0	0	0	0		

(2,2) - Iteration 2

aa = 2
bb = 2

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	11	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	0	0	0	10	1
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	1	0	0	00	1
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	1	00	1
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	1	0	0	1	01	2
0x0040	...	0	0	0	0		

(2,2) - Iteration 3

aa = 3
bb = 4

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	01	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	1	0	0	11	2
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	1	0	1	11	2
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	2	11	2
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	1	0	0	2	11	3
0x0040	...	0	0	0	0		

(2,2) - Iteration 4

aa = 2
bb = 2

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	11	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	1	0	0	10	2
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	1	0	1	00	2
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	2	00	2
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	2	0	0	2	01	4
0x0040	...	0	0	0	0		

(2,2) - Iteration 5

aa = 3
bb = 4

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	01	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	2	0	0	11	3
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	1	0	2	11	3
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	3	11	2
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	2	0	0	3	11	4
0x0040	...	0	0	0	0		

(2,2) - Iteration 6

aa = 2
bb = 2

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	11	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	2	0	0	10	3
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	1	0	2	00	3
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	3	00	2
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	3	0	0	3	01	4
0x0040	...	0	0	0	0		

(2,2) - Iteration 7

aa = 3
bb = 4

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	01	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	3	0	0	11	3
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	1	0	3	11	3
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	3	11	2
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	3	0	0	3	11	4
0x0040	...	0	0	0	0		

(2,2) - Iteration 8

aa = 2
bb = 2

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	11	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	3	0	0	10	3
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	1	0	3	00	3
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	3	00	2
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	3	0	0	3	01	4
0x0040	...	0	0	0	0		

(2,2) - Iteration 100

aa = 2
bb = 2

address	Instruction	00	01	10	11	2-b sft reg	misP. counter
0x0000	L0: ...	0	0	0	0	11	
...	...	0	0	0	0		
0x0010	DADDUI R3, R1, #-2	0	0	0	0		
0x0014	BNEZ R3, L1	1	3	0	0	10	3
0x0018	DADD R1, R0, R0	0	0	0	0		
0x001C	L1: DADDUI R3, R2, #-2	0	0	0	0		
0x0020	BNEZ R3, L2	0	1	0	3	00	3
0x0024	DADD R2, R0, R0	0	0	0	0		
0x0028	L2: DSUB R3, R1, R2	0	0	0	0		
0x002C	BNEZ R3, L3	0	0	0	3	00	2
0x0030	...	0	0	0	0		
0x0034	L3: ...	0	0	0	0		
0x0038	DADDI R4, R4, #-1	0	0	0	0		
0x003C	BNEZ R4, L0	2	0	0	3	00	5
0x0040	...	0	0	0	0		

(2,2) Example

Considering a (2,2) correlating predictor of 1K entries; and assuming that the processor executes the following code fragment, determine the (2,2) final state and calculate the misprediction ratio for the presented case.

Case 1:

-Program inputs for aa and bb alternate values different than 2 and equal to 2.

Misprediction ratio =

Number of mispredicted branches / Total branches

Mispredition ratio = 13 / 400

Mispredition ratio = 3.25%